Introduction to Mathematics for Deep Learning: Midterm

Tuesday, October 21, 2025

- You can use previous parts of a problem even if you did not solve them.
- As throughout the course, entropy (H) and Mutual Information (I) are in bits.
- log is taken in base 2.

ID : _____

Name : _____

1	/ 50
2	/ 25
3	/ 25
Total	/ 100

extra sheet

1. **Mix of Questions** *(50 points, 10 points each)*

1) Here is a statement about pairwise independence and joint independence. Let X, Y_1, and Y_2 be binary random variables. If $I(X;Y_1) = 0$ and $I(X;Y_2) = 0$, does it follow that $I(X;Y_1,Y_2) = 0$? Prove or provide a counterexample.

2) Suppose a (non-stationary) Markov chain starts in one of n states, necks down to $k < n$ states, and then fans back to $m > k$ states. Thus $X_1 - X_2 - X_3$ forms a Markov chain, where $X_1 \in \{1, 2, \ldots, n\}$, $X_2 \in \{1, 2, \ldots, k\}$, $X_3 \in \{1, 2, \ldots, m\}$, and $p(x_1, x_2, x_3) = p(x_1)p(x_2|x_1)p(x_3|x_2)$. Show that the dependence of X_1 and X_3 is limited by the bottleneck by proving that $I(X_1; X_3) \leq \log k$.

3) For independent continuous random variables X and Y,
show that $h(X + Y) \geq h(X)$. [hint: Consider $h(X + Y|Y)$]

4) The *World Series* is a seven-game series that terminates as soon as either team wins four games. Let X be the random variable representing the full outcome of the series between teams A and B; for example, possible values of X include sequences such as AAAA, BABABAB, and BBBAAAA. Assuming that: i) the two teams are equally matched (each game is won by A or B with probability 1/2), and ii) all games are independent. Compute $H(X)$.

5) Let $\{X_t\}_{t\geq 0}$ be a 1-st order Markov process on the finite state space $\{0, 1, \ldots, K\}$ with transition probabilities

$$P(X_{t+1} = j | X_t = i) = \begin{cases} 1 - \varepsilon & \text{if } i = j \geq 1 \\ \varepsilon & \text{if } j = 0 \text{ and } i \geq 1 \\ 1 & \text{if } i = j = 0 \\ 0 & \text{otherwise} \end{cases}$$

Assume $0 < \varepsilon < 1$. Compute the *limit distribution* of the process, i.e., π_∞.

problem 1 cont.

2. **MLE** *(25 points)*

Let $X_1, \ldots, X_n$ be i.i.d. samples from an exponential distribution with parameter $\lambda > 0$, i.e.,

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

(a) Derive the maximum likelihood estimator (MLE) $\widehat{\lambda}_{\text{MLE}}$ of λ. *(10 points)*

(b) Show that $\mathbb{E}[1/\widehat{\lambda}_{\text{MLE}}] = 1/\lambda$. *(5 points)*

(c) Show that $\text{Bias}(\widehat{\lambda}_{\text{MLE}}) > 0$. *(10 points)*

problem 2 cont.

3. **JS-Divergence** *(25 points)*

Let P and Q be discrete probabilities on the alphabet $\{1, 2, \cdots, K\}$. Relative entropy $D(P||Q)$, can be viewed as a distance between probability distributions. In general it is not symmetric, i.e., there exist P and Q for which $D(P||Q) \neq D(Q||P)$. We can construct symmetric distance measures between two distributions based on relative entropy; one such measure is called the Jensen-Shannon Divergence. For $R = (P + Q)/2$, i.e., R is a new probability distribution on the alphabet $\{1, 2, \cdots, K\}$ such that

$$R(i) = \frac{P(i) + Q(i)}{2} \quad \text{for all } i \in \{1, \ldots, K\}$$

the JS-Divergence is defined as

$$JS(P||Q) = \frac{1}{2} D\left(P \| R\right) + \frac{1}{2} D\left(Q \| R\right).$$

It is easy to check that $JS(P||Q) = JS(Q||P)$.

(a) Let Z be a Bernoulli($\frac{1}{2}$) random variable. Let X be a random variable such that $X|\{Z = 1\}$ has a distribution P while $X|\{Z = 0\}$ has a distribution Q. Find the pmf p_X of random variable X as a function of P, Q, R. *(5 points)*

(b) For the same X and Z in the previous problem, express $JS(P\|Q)$ in terms of $I(X; Z)$. *(10 points)*

(c) What is the maximum value of $JS(P||Q)$ among all possible distributions P and Q. [Hint: Use the previous results.] *(10 points)*

problem 3 cont.

extrasheet.

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